Marcel **Walter**

Postdoctoral Researcher · Senior Quantum Software Engineer

Munich, Germany

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Education

Technical University of Munich (TUM) *Munich, Germany*

Habilitation Candidate *2026 – Present*

- Thesis: *Advanced Design Automation and Simulation for Atomic-Scale Computing*

University of Bremen *Bremen, Germany*

Ph.D. in Computer Science (summa cum laude) *2016 – 2021*

- Thesis: *Design Automation for Field-coupled Nanotechnologies*

University of Bremen *Bremen, Germany*

M.Sc. in Computer Science *2015 – 2016*

- Specialization in *Safety, Security, and Quality*
- Thesis: *Exact Synthesis of Abstraction-Level-Independent Logic Networks*

University of Bremen *Bremen, Germany*

B.Sc. in Computer Science *2011 – 2015*

- Thesis: *Optimization of Quantum Circuits for Linear Nearest Neighbor Architectures*

Experience

Munich Quantum Software Company (MQSC) *Munich, Germany*

Senior Quantum Software Engineer *2025 – Present*

- Industrial-strength software for quantum computers
- HPCQC integration

University of Bremen *Bremen, Germany*

Substitute Professor *2024*

- Teaching graduate courses concerning computational nanotechnology, logic synthesis, and physical design

Technical University of Munich (TUM) *Munich, Germany*

Postdoctoral Researcher *2021 – Present*

- Design Automation for post-CMOS Emerging Nanotechnologies
- Logic Synthesis & Physical Design
- Silicon Photonics
- Superconducting Quantum Computing
- Advising theses of all levels
- Teaching undergraduate and graduate courses
- *Supervisor:* Prof. Dr. Robert Wille, Chair for Design Automation

University of Bremen *Bremen, Germany*

Researcher and Ph.D. Student *2016 – 2021*

- Physical design of post-CMOS Emerging Nanotechnologies
- Verification of CRAM plans for PR2 robots at the collaborative research center EASE
- Teaching undergraduate courses and supervising Bachelor's and Master's theses
- *Supervisor:* Prof. Dr. Rolf Drechsler, Research Group of Computer Architecture

University of Bremen *Bremen, Germany*

Independent Study *2016*

- Reactive system synthesis from LTL specifications using Satisfiability Modulo Theories (SMT)
- *Supervisor:* Prof. Dr. Rüdiger Ehlers, Modelling of Technical Systems Research Group

German Research Center for Artificial Intelligence (DFKI) *Bremen, Germany*

Research Assistant *2015 – 2016*

- Formal verification of access control systems
- *Supervisor:* Prof. Dr. Christoph Lüth, Cyber Physical Systems

University of Bremen
Bremen, Germany

Student Tutor / Teaching Assistant — *2013 – 2015*

- Practical Computer Science 1 (Java basics, OOP, algorithms and data structures)
- *Supervisor:* Dr. Karsten Hölscher, Software Engineering Group
- Technical Computer Science 1 (Computer architecture, Boolean Algebra, logic synthesis)
- *Supervisor:* Prof. Dr. Rolf Drechsler, Research Group of Computer Architecture

Research Interests

- **Design Automation for Post-CMOS Nanotechnologies:** Field-coupled Nanocomputing (Quantum-dot Cellular Automata, Silicon Dangling Bonds) — logic synthesis, physical design, placement & routing, verification, and physical simulation.
- **Quantum Computing:** software and hardware integration for quantum computers, including cryogenic CMOS control electronics and HPC-quantum computing (HPCQC) integration.
- **Emerging Substrates:** silicon photonics and superconducting quantum circuit design, extending the same design-automation methodology to new physical substrates.
- **Algorithmic Foundations:** exact and heuristic combinatorial optimization, SAT/SMT-based synthesis and verification, graph algorithms, and computational complexity theory.

Languages

German (Native) · English (Fluent) · Dutch (B2)

Honors & Awards

2025	**Best Student Paper Award**, IEEE NANO 2025
2025	**Best Paper Award Candidate**, LASCAS 2025
2024	**Teaching Award Candidate (*Studierendenpreis*)**, Berninghausenpreis 2024
2024	**Best Presentation Award**, IEEE NANO 2024
2023	**Best Paper Award Candidate**, IWLS 2023
2021	**Best Paper Award Candidate**, ASP-DAC 2021
2020	**Best Young Student Fellow Research Video Award**, DAC 2020
2020	**Young Student Fellow Program Scholarship**, DAC 2020
2020	**Best Student Forum Paper Award**, ISVLSI 2020
2020	**Best Paper Award Candidate**, ISVLSI 2020
2019	**Best Research Demo Award**, ISVLSI 2019

Talks

Ningbo University
Ningbo, China

Invited Talk — *Jul. 2026*

- From Logic to Atoms: Design Automation for Atomic-Scale Computing

Free Silicon Conference
Frankfurt (Oder), Germany

Invited Talk — *Jul. 2025*

- aigverse: Toward Machine Learning-Driven Logic Synthesis

4th Annual Hack4Her Event
Amsterdam, Netherlands

Workshop with Hands-on — *Jun. 2025*

- Chip Design for AI

Universidade Federal de Minas Gerais - Workshop on Nanocomputing
Belo Horizonte, Brazil

Invited Workshop Contribution — *Feb. 2025*

- The Munich Nanotech Toolkit (MNT)

IEEE International Conference on Nanotechnology (IEEE NANO) 2024
Gijón, Spain

Special Session Host and Organizer — *Jul. 2024*

- Field-coupled Nanocomputing (FCN)

3rd Annual Hack4Her Event
Amsterdam, Netherlands

Workshop with Hands-on — *Jun. 2024*

- Logic Synthesis for AI

International Workshop on Logic and Synthesis (IWLS) 2024 — *Zurich, Switzerland*

INVITED SPECIAL SESSION — *Jun. 2024*

- The Case for Planar Logic Synthesis – Crossing Costs in Nanotech

University of Alberta — *Edmonton, Canada*

GUEST RESEARCHER LECTURE — *Mar. 2024*

- The Munich Nanotech Toolkit (MNT)

EPFL - Laboratoire des Systèmes Intégrés — *Lausanne, Switzerland*

THESIS OVERVIEW — *May 2021*

- Design Automation for Field-coupled Nanotechnologies

Johannes Kepler University (JKU) Linz — *Linz, Austria*

GUEST LECTURE — *Jan. 2019*

- Field-coupled Nanocomputing (FCN)

Teaching Experience

University of Bremen — *Bremen, Germany*

GRADUATE COURSES — *2024*

- Logic Synthesis & Physical Design
- Unconventional Computing: Nanotech and Biochips
- Seminar: Computational Nanotechnologies

Technical University of Munich (TUM) — *Munich, Germany*

UNDERGRADUATE & GRADUATE COURSES — *2022 – Present*

- Introduction to Emerging Computing Technologies
- Logic Synthesis & Physical Design

University of Bremen — *Bremen, Germany*

ADVANCED TOPICS IN TECHNICAL COMPUTER SCIENCE (SEMINAR) — *2017 – 2020*

- Co-led with Prof. Dr. Rolf Drechsler; Boolean Algebra, Logic Synthesis, Graph Theory, hardware description languages, and post-CMOS emerging technologies

University of Bremen — *Bremen, Germany*

HEAD OF ORGANIZATION, TECHNICAL COMPUTER SCIENCE 1 — *2017 – 2018*

- Lecture hall management, briefing tutors, addressing student concerns, creating and correcting assignment sheets, proxy lectures, and organizing exams

University of Bremen — *Bremen, Germany*

UNDERGRADUATE COURSES — *2013 – 2021*

- Technical Computer Science 1 (Computer Architecture, Logic Synthesis)
- Practical Computer Science 1 (Java, Algorithms and Data Structures)

Supervision & Mentoring

PH.D. CANDIDATES

Technical University of Munich (TUM) — *Munich, Germany*

JAN DREWNIOK — PH.D. (*summa cum laude*) — *2026*

- Thesis: Design and Simulation of Atomic-Scale Computing: A Comprehensive Framework for Physical Simulation, Logic Design, Logic Analysis, and Circuit Design
- IEEE Nanotechnology Council (NTC) 2026 Best PhD Thesis Award
- DATE 2025 Best Poster Award at the PhD Forum
- IEEE NANO 2024 Student Design Competition Winner: *NanoLogic: An Educational App for Atomic-Scale Computing*

SIMON HOFMANN — PH.D. (*summa cum laude*) — *2026*

- Thesis: Physical Design for Field-coupled Nanocomputing
- LASCAS 2025 Best Paper Award Candidate: *Physical Design for Field-coupled Nanocomputing with Discretionary Cost Objectives*

Bachelor's & Master's Theses (Selection)

2025	**Michael Feldmeier (M.Sc.)**, Exploring Feedback-Guided AIG Optimization to Align Logic Synthesis with Mapping Objectives	*TUM*
2025	**Danilo Quinci (M.Sc.)**, Design of Standard Cell Libraries for Molecular Field-coupled Nanocomputing Design Automation	*PoliTo*
2025	**Felix Kiefhaber (B.Sc.)**, Super-Tile Routing for Omnidirectional Information Flow in Silicon Dangling Bond Logic	*TUM*
2024	**Willem Lambooy (M.Sc.)**, Hierarchically Bounded Composable Interactions: Applications and Extensions to Atom-Scale Logic Design	*Radboud*
2023	**Benjamin Hien (M.Sc.)**, Signal Distribution Networks in Automatic QCA Standard-Cell Placement & Routing	*TUM*
2023	**Isabella Venancia Gardner (B.Sc.)**, Machine Learning for Electrostatic Ground State Simulation of Silicon Dangling Bond Logic	*VU Amsterdam*
2021	**Sophia Kuhn (B.Sc.)**, Design Understanding für Quantum-dot Cellular Automata Layouts	*Uni Bremen*
2021	**Daniel Staack (B.Sc.)**, Fehlerfindung in robotischen Plänen mittels Fuzzing	*Uni Bremen*
2020	**Till Schlechtweg (B.Sc.)**, Gate-level Placement für Field-coupled Nanocomputing unter Berücksichtigung von Clocking Constraints	*Uni Bremen*

Open-Source Projects (Selection)

Munich Nanotech Toolkit (MNT)
https://github.com/cda-tum/fiction

Initiator and Core Maintainer — *2018 – Present*

- Initiated and maintain the Munich Nanotech Toolkit, an ecosystem of open-source tools for design automation of post-CMOS nanotechnologies (cf. the tool paper [46]), contributing across its components as needed.
- Flagship project: fiction, a framework for Design Automation of Field-coupled Nanotechnologies for researchers and developers (cf. the tool paper [83]), honored with the ISVLSI 2019 Best Research Demo Award.

aigverse
https://github.com/marcelwa/aigverse

Initiator and Maintainer — *2024 – Present*

- A Python library for working with logic networks, synthesis, and optimization.
- High-performance C++ backend and interoperability with other logic synthesis tools.
- Integration with popular data science and machine learning libraries.

Munich Quantum Toolkit (MQT)
https://github.com/munich-quantum-toolkit/core

Contributor — *2025 – Present*

- Contributing across repositories of the Munich Quantum Toolkit as part of ongoing quantum software engineering work at MQSC.
- Exemplary repository: MQT Core, the toolkit's foundational C++/Python library providing the shared data structures and algorithms (compilation, simulation, verification) that its other quantum-software components build on (cf. the tool paper).

EPFL Logic Synthesis Libraries
https://github.com/lsils/mockturtle

Contributor — *2020 – Present*

- Contributing to the EPFL Logic Synthesis Libraries, a collection of C++ libraries for logic network synthesis, optimization, and verification.
- Exemplary repository: mockturtle, a header-only library for logic network manipulation and optimization.

QDMI-on-IQM
https://github.com/iqm-finland/QDMI-on-IQM

Contributor and co-Maintainer — *2025 – Present*

- Implementation of the Quantum Device Management Interface (QDMI) for IQM's quantum computing hardware, the standard interface connecting HPC software stacks to quantum backends.
- Enables the HPCQC integration case study demonstrated on real IQM hardware [81].
- Co-maintain the repository: reviewing contributions and keeping the integration current with both QDMI and IQM's hardware interfaces.

Professional Service

Executive Team and Organizing Committees

- 2027 **TPC Member**, IEEE International Conference on Nanotechnology (IEEE NANO)
- 2027 **Program co-Chair and TPC Member**, International Workshop on Logic and Synthesis (IWLS)
- 2026 **Best Poster Award Judge**, IEEE International Conference on Nanotechnology (IEEE NANO)
- 2026 **Publicity co-Chair, TPC Member, and Session Chair**, International Workshop on Logic and Synthesis (IWLS)
- 2025 **Track Co-Chair**, IEEE CASS Latin American Symposium on Circuits and Systems (LASCAS)
- 2024 **Proceedings Chair, TPC Member, and Session Chair**, International Workshop on Logic and Synthesis (IWLS)
- 2024 **Special Session Host and Organizer**, IEEE International Conference on Nanotechnology (IEEE NANO)
- 2023 **TPC Member**, International Conference on Computer-Aided Design (ICCAD)
- 2023 **Session Chair**, International Workshop on Logic and Synthesis (IWLS)
- 2020 **Technical Operator**, International Conference on Computer-Aided Design (ICCAD)
- 2018 **Staff Member**, European Test Symposium (ETS)

Reviewer for International Conferences and Journals

- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- Design Automation Conference (DAC)
- International Conference On Computer Aided Design (ICCAD)
- Design, Automation and Test in Europe (DATE)
- ACM Transactions on Design Automation of Electronic Systems (TODAES)
- ACM Journal on Emerging Technologies in Computing (JETC)
- IEEE Transactions on Emerging Topics in Computing (TETC)
- IEEE Transactions on Nanotechnology (TNANO)
- IEEE Transactions on Circuits and Systems (TCAS)
- IEEE Transactions on Very Large Scale Integration Systems (TVLSI)
- Journal of Computational Electronics (JCEL)
- International Workshop on Logic and Synthesis (IWLS)
- ACM Great Lakes Symposium on VLSI (GLSVLSI)
- Euromicro Conference on Digital System Design (DSD)
- Microprocessors and Microsystems (MICPRO)
- Analog Integrated Circuits and Signal Processing (ALOG)
- International Conference on VLSI Design (VLSID)
- European Test Symposium (ETS)
- IEEE Design & Test (D&T)
- IEEE International Symposium on Multiple-Valued Logic (ISMVL)
- IEEE International Symposium on Embedded Computing and System Design (ISED)
- IEEE CASS Latin American Symposium on Circuits and Systems (LASCAS)
- Mathematics
- EPJ Plus
- Integration - The VLSI Journal
- IEEE Embedded Systems Letters (ESL)

Community Service

- 2022–24 **Expert and Mentor**, Hack4Her Event / Technology Retreat / Hackathon — *Amsterdam, Netherlands*
- 2013 – Present **Student Mentor**, Gymnasium Syke — *Syke, Germany*

Challenges, Tutorials & Contests

ISPD Contest

Participant and Mentor — *2026*

- Post-Placement Buffering and Sizing.

IWLS Programming Contests

Participant and Mentor — *2024*

- Optimal synthesis of AIGs and XAIGs from truth table specifications.

High Level Synthesis Tutorial by Cadence @ DAC 2020

Participant — *2020*

- Logic synthesis of a CNN AI Accelerator for Image Recognition.

Bremen Big Data Challenge

Participant — *2016*

- Applied machine learning on League of Legends statistics to predict winning teams.

Publications

Books

[1] **M. Walter**, R. Wille, F. Sill Torres, R. Drechsler, *Design Automation for Field-coupled Nanotechnologies*. Springer, 2022. DOI: 10.1007/978-3-030-89952-3

Journal Articles

[2] S. S. H. Ng, **M. Walter**, S. Hofmann, J. Drewniok, R. Wille, K. Walus, "RTL-to-Atoms Synthesis of a Machine Learning Accelerator on Atomic-Scale Computers," *IEEE Transactions on Nanotechnology (TNANO)*, 2026. DOI: 10.1109/TNANO.2026.3689032

[3] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, "QuickCell: Fast Automatic Design of Standard Cells for Silicon Dangling Bond Logic," *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. DOI: 10.1109/TCAD.2025.3605537

[4] S. Hofmann, **M. Walter**, R. Wille, "Efficient and Scalable Post-Layout Optimization for Field-coupled Nanotechnologies," *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. DOI: 10.1109/TCAD.2025.3549354

[5] S. Hofmann, **M. Walter**, R. Wille, "Graph-Oriented Layout Design for Field-coupled Nanocomputing via Parallel Multi-Objective Search Space Exploration," *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)*, 2025. DOI: 10.1109/TCSI.2025.3614709

[6] S. S. H. Ng, J. Croshaw, **M. Walter**, R. Wille, R. Wolkow, K. Walus, "Simulating Charged Defects in Silicon Dangling Bond Logic Systems to Evaluate Logic Robustness," *IEEE Transactions on Nanotechnology (TNANO)*, 2024. DOI: 10.1109/TNANO.2024.3372946

[7] M. D. Vieira, S. S. H. Ng, **M. Walter**, R. Wille, K. Walus, R. S. Ferreira, O. P. Vilela Neto, J. A. M. Nacif, "Three-Input NPN Class Gate Library for Atomic Silicon Quantum Dots," *IEEE Design & Test*, 2022. DOI: 10.1109/MDAT.2022.3189814

[8] F. Sill Torres, P. A. Silva, G. Fontes, **M. Walter**, J. A. M. Nacif, R. S. Ferreira, O. P. V. Neto, J. F. Chaves, R. Wille, P. Niemann, D. Große, R. Drechsler, "On the Impact of the Synchronization Constraint and Interconnections in Quantum-dot Cellular Automata," *Microprocessors and Microsystems (MICPRO)*, 2020. DOI: 10.1016/j.micpro.2020.103109

[9] **M. Walter**, R. Wille, D. Große, F. Sill Torres, R. Drechsler, "Placement and Routing for Tile-based Field-coupled Nanocomputing Circuits is NP-complete (Research Note)," *Journal on Emerging Technologies in Computing Systems (JETC)*, 2019. DOI: 10.1145/3312661

Conference Proceedings

[10] J. Drewniok, **M. Walter**, R. Wille, "Operational Domain Sketching for Silicon Dangling Bond Logic," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.

[11] M. Feldmeier, **M. Walter**, R. Wille, "A Framework for Post-Mapping Feedback-Guided Selective Logic Transformations," in *International Workshop on Logic Synthesis (IWLS)*, 2026.

[12] I. V. Gardner, **M. Walter**, R. Wille, M. Cochez, "AIG2PT: A Generative Pre-trained Transformer for Unconditional And-Inverter Graph Synthesis," in *International Workshop on Logic Synthesis (IWLS)*, 2026.

[13] B. Hien, **M. Walter**, R. Wille, "Minimizing Area and Delay in Planar Physical Design for Field-coupled Nanocomputing," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.

[14] B. Hien, **M. Walter**, R. Wille, "Toward Compact Planar Circuits for Atomic-Scale Computing: A Hybrid Crossing Elimination Flow," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.

[15] W. Lambooy, J. Drewniok, **M. Walter**, R. Wille, "Holistic Physical Design for Silicon Atomic Logic: Cramming More Components onto Clock Zones," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.

[16] W. Lambooy, J. Drewniok, **M. Walter**, R. Wille, "Mastering the Exponential Complexity of Exact Physical Simulation of Silicon Dangling Bonds," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2026.

[17] S. S. H. Ng, **M. Walter**, S. Hofmann, J. Drewniok, R. Wille, K. Walus, "Design and Emulation Methodology for Atomic-Scale Systolic Arrays: An LLM Accelerator Case Study in Silicon DB Logic," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.

[18] S. S. H. Ng, **M. Walter**, M. Yuan, J. Drewniok, R. H. Foote, R. Lupoiu, J. A. Fan, R. Wolkow, R. Wille, K. Walus, "BitPlanarNet: Learning-to-Layout Planar Gate Networks for Emerging Devices," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.

[19] **M. Walter**, M. Feldmeier, R. Wille, “Exact Synthesis with Optimal Switching Activity,” in *Design, Automation and Test in Europe (DATE)*, 2026.

[20] **M. Walter**, I. V. Gardner, R. Wille, “aigverse: A Unified Infrastructure for Machine Learning-Driven Logic Synthesis,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.

[21] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “Towards Fast Automatic Design of Silicon Dangling Bond Logic,” in *Design, Automation and Test in Europe (DATE)*, 2025.

[22] J. Drewniok, **M. Walter**, R. Wille, “QuickTrace: An Efficient Contour Tracing Algorithm for Defect Robustness Simulation of Silicon Dangling Bond Logic,” in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025.

[23] I. V. Gardner, **M. Walter**, Y. Miyasaka, R. Wille, M. Cochez, “Bias by Design: Diversity Quantification to Mitigate Structural Bias Effects in AIG Logic Optimization,” in *Design, Automation and Test in Europe (DATE)*, 2025. DOI: `10.23919/DATE64628.2025.10993009`

[24] B. Hien, D. Quinci, Y. Ardesi, G. Beretta, F. Ravera, **M. Walter**, R. Wille, “Bridging the Gap Between Molecular FCN and Design Automation with SIM(7)-MolPDK: A Physically Simulated Standard-Cell Library,” in *IEEE Latin American Conference on Nanotechnology (IEEE LANANO)*, 2025.

[25] B. Hien, **M. Walter**, A. Tempia Calvino, A. Mishchenko, R. Wille, “Ashenhurst-Curtis Decomposition Using Don’t Cares,” in *International Workshop on Logic Synthesis (IWLS)*, 2025.

[26] B. Hien, **M. Walter**, S. Hofmann, R. Wille, “A Fully Planar Approach to Field-coupled Nanocomputing: Scalable Placement and Routing Without Wire Crossings,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025.

[27] S. Hofmann, J. Drewniok, **M. Walter**, R. Wille, “MNT Designer: A Comprehensive Design Tool for Field-coupled Nanocomputing,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025.

[28] S. Hofmann, **M. Walter**, R. Wille, “Late Breaking Results: Physical Co-Design for Field-coupled Nanocomputing,” in *Design, Automation and Test in Europe (DATE)*, 2025.

[29] S. Hofmann, **M. Walter**, R. Wille, “Physical Design for Field-coupled Nanocomputing with Discretionary Cost Objectives,” in *IEEE Latin American Symposium on Circuits and Systems (LASCAS)*, **Best Paper Award Candidate**, 2025.

[30] S. S. H. Ng, **M. Walter**, J. Drewniok, S. Hofmann, R. Wille, K. Walus, “Building a Machine Learning Accelerator with Silicon Dangling Bonds: From Verilog to Quantum Dot Layout,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, **Best Student Paper Award**, 2025.

[31] **M. Walter**, J. Drewniok, R. Wille, “Live Demonstration: An Application for Layout Resilience Analysis of Silicon Dangling Bond Logic,” in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025.

[32] **M. Walter**, J. Drewniok, R. Wille, “The Operational Domain Explorer: A Comprehensive Framework to Unveil the Thermal Landscape of Silicon Dangling Bond Logic Beyond Conventional Operability,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025.

[33] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “On-the-fly Defect-Aware Design of Circuits based on Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: `10.1109/NANO61778.2024.10628962`

[34] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “Unifying Figures of Merit: A Versatile Cost Function for Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: `10.1109/NANO61778.2024.10628671`

[35] J. Drewniok, **M. Walter**, R. Wille, “The Need for Speed: Efficient Exact Simulation of Silicon Dangling Bond Logic,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024. DOI: `10.1109/ASP-DAC58780.2024.10473946`

[36] B. Hien, **M. Walter**, V. M. Santen, F. Klemme, S. S. Parihar, G. Pahwa, Y. S. Chauhan, H. Amrouch, R. Wille, “Technology Mapping for Cryogenic CMOS Circuits,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2024. DOI: `10.1109/ISVLSI61997.2024.00057`

[37] B. Hien, **M. Walter**, R. Wille, “Reducing Wire Crossings in Field-Coupled Nanotechnologies,” in *International Workshop on Logic Synthesis (IWLS)*, 2024.

[38] B. Hien, **M. Walter**, R. Wille, “Reducing Wire Crossings in Field-Coupled Nanotechnologies,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: `10.1109/NANO61778.2024.10628717`

[39] S. Hofmann, **M. Walter**, L. Servadei, R. Wille, “Thinking Outside the Clock: Physical Design for Field-coupled Nanocomputing with Deep Reinforcement Learning,” in *International Symposium on Quality Electronic Design (ISQED)*, 2024. DOI: `10.1109/ISQED60706.2024.10528711`

[40] S. Hofmann, **M. Walter**, R. Wille, “A* is Born: Efficient and Scalable Physical Design for Field-coupled Nanocomputing,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628808
[41] S. Hofmann, **M. Walter**, R. Wille, “Late Breaking Results: Wiring Reduction for Field-coupled Nanotechnologies,” in *Design Automation Conference (DAC)*, 2024. DOI: 10.1145/3649329.3663491
[42] S. Hofmann, **M. Walter**, R. Wille, “MNT Bench: Benchmarking Software and Layout Libraries for Field-coupled Nanocomputing,” in *Design, Automation and Test in Europe (DATE)*, 2024. DOI: 10.23919/DATE58400.2024.10546596
[43] D. Marakkalage, **M. Walter**, S.-Y. Lee, R. Wille, G. De Micheli, “Technology Mapping for Beyond-CMOS Circuitry with Unconventional Cost Functions,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, **Best Presentation Award**, 2024. DOI: 10.1109/NANO61778.2024.10628909
[44] S. S. H. Ng, J. Drewniok, **M. Walter**, J. Retallick, R. Wille, K. Walus, “Unlocking Flexible Silicon Dangling Bond Logic Designs on Alternative Silicon Orientations,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628802
[45] **M. Walter**, J. Croshaw, S. S. H. Ng, K. Walus, R. Wolkow, R. Wille, “Towards Atomic Defect-Aware Physical Design of Silicon Dangling Bond Logic on the H-Si(100)-2x1 Surface,” in *Design, Automation and Test in Europe (DATE)*, 2024. DOI: 10.23919/DATE58400.2024.10546683
[46] **M. Walter**, J. Drewniok, S. Hofmann, B. Hien, R. Wille, “The Munich Nanotech Toolkit (MNT),” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628747
[47] **M. Walter**, J. Drewniok, R. Wille, “Ending the Tyranny of the Clock: SAT-based Clock Number Assignment for Field-coupled Nanotechnologies,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628908
[48] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “QuickSim: Efficient and Accurate Physical Simulation of Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2023. DOI: 10.1109/NANO58406.2023.10231266
[49] J. Drewniok, **M. Walter**, R. Wille, “Minimal Design of SiDB Gates: An Optimal Basis for Circuits Based on Silicon Dangling Bonds,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2023. DOI: 10.1145/3611315.3633241
[50] J. Drewniok, **M. Walter**, R. Wille, “Temperature Behavior of Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2023. DOI: 10.1109/NANO58406.2023.10231259
[51] S. Hofmann, **M. Walter**, L. Servadei, R. Wille, “Late Breaking Results From Hybrid Design Automation for Field-coupled Nanotechnologies,” in *Design Automation Conference (DAC)*, 2023. DOI: 10.1109/DAC56929.2023.10247933
[52] S. Hofmann, **M. Walter**, L. Servadei, R. Wille, “Thinking Outside the Clock: Physical Design for Field-coupled Nanocomputing with Deep Reinforcement Learning,” in *International Workshop on Logic Synthesis (IWLS)*, 2023.
[53] S. Hofmann, **M. Walter**, R. Wille, “Post-Layout Optimization for Field-coupled Nanotechnologies,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2023. DOI: 10.1145/3611315.3633247
[54] S. Hofmann, **M. Walter**, R. Wille, “Scalable Physical Design for Silicon Dangling Bond Logic: How a 45°~Turn Prevents the Reinvention of the Wheel,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2023. DOI: 10.1109/NANO58406.2023.10231278
[55] D. Marakkalage, **M. Walter**, S.-Y. Lee, R. Wille, G. De Micheli, “Technology Mapping for Beyond-CMOS Circuitry with Unconventional Cost Functions,” in *International Workshop on Logic Synthesis (IWLS)*, **Best Paper Award Candidate**, 2023.
[56] V. M. Santen, **M. Walter**, F. Klemme, S. S. Parihar, G. Pahwa, Y. Chauhan, R. Wille, H. Amrouch, “Design Automation for Cryogenic CMOS Circuits,” in *International Workshop on Logic Synthesis (IWLS)*, 2023.
[57] V. M. Santen, **M. Walter**, F. Klemme, S. S. Parihar, G. Pahwa, Y. Chauhan, R. Wille, H. Amrouch, “Design Automation for Cryogenic CMOS Circuits,” in *Design Automation Conference (DAC)*, 2023. DOI: 10.1109/DAC56929.2023.10247824
[58] **M. Walter**, J. Drewniok, S. S. H. Ng, K. Walus, R. Wille, “Reducing the Complexity of Operational Domain Computation in Silicon Dangling Bond Logic,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2023. DOI: 10.1145/3611315.3633246
[59] **M. Walter**, B. Hien, R. Wille, “Versatile Signal Distribution Networks for Scalable Placement and Routing of Field-coupled Nanocomputing Technologies,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2023. DOI: 10.1109/ISVLSI59464.2023.10238604

[60] W. Lambooy, **M. Walter**, R. Wille, “Exploiting the Third Dimension: Stackable Quantum-dot Cellular Automata,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2022. DOI: 10.1145/3565478.3572529
[61] **M. Walter**, S. S. Hang Ng, K. Walus, R. Wille, “Hexagons are the Bestagons: Design Automation for Silicon Dangling Bond Logic,” in *Design Automation Conference (DAC)*, 2022. DOI: 10.1145/3489517.3530525
[62] **M. Walter**, R. Wille, “Efficient Multi-Path Signal Routing for Field-coupled Nanotechnologies,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2022. DOI: 10.1145/3565478.3572539
[63] **M. Walter**, W. Haaswijk, R. Wille, F. Sill Torres, R. Drechsler, “One-Pass Synthesis for Field-coupled Nanocomputing Technologies,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, **Best Paper Award Candidate**, 2021. DOI: 10.1145/3394885.3431607
[64] U. Garlando, **M. Walter**, R. Wille, F. Riente, F. Sill Torres, R. Drechsler, “ToPoliNano and fiction: Design Tools for Field-coupled Nanocomputing,” in *Euromicro Conference on Digital System Design (DSD)*, 2020. DOI: 10.1109/DSD51259.2020.00071
[65] T. Meywerk, **M. Walter**, D. Große, R. Drechsler, “Clustering-Guided SMT(LRA) Learning,” in *International Conference on Integrated Formal Methods (IFM)*, 2020. DOI: 10.1007/978-3-030-63461-2_3
[66] T. Meywerk, **M. Walter**, V. Herdt, J. Kleinekathöfer, D. Große, R. Drechsler, “Verifying Safety Properties of Robotic Plans Operating in Real-World Environments via Logic-Based Environment Modeling,” in *International Symposium on Leveraging Applications of Formal Methods (ISoLA)*, 2020. DOI: 10.1007/978-3-030-61470-6_20
[67] **M. Walter**, R. Drechsler, “Design Automation for Field-coupled Nanotechnologies,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, **Best Student Forum Paper Award**, 2020, pp. 176–181. DOI: 10.1109/ISVLSI49217.2020.00040
[68] **M. Walter**, W. Haaswijk, R. Wille, F. Sill Torres, R. Drechsler, “SAT-based Exact Physical Design for Field-coupled Nanocomputing Technologies,” in *International Workshop on Logic Synthesis (IWLS)*, 2020.
[69] **M. Walter**, R. Wille, F. Sill Torres, R. Drechsler, “Bail on Balancing: An Alternative Approach to the Physical Design of Field-coupled Nanocomputing Circuits,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, **Best Paper Award Candidate**, 2020. DOI: 10.1109/ISVLSI49217.2020.00022
[70] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “Verification for Field-coupled Nanocomputing Circuits,” in *Design Automation Conference (DAC)*, 2020. DOI: 10.5555/3437539.3437614
[71] T. Meywerk, **M. Walter**, V. Herdt, D. Große, R. Drechsler, “Towards Formal Verification of Plans for Cognition-enabled Autonomous Robotic Agents,” in *Euromicro Conference on Digital System Design (DSD)*, 2019. DOI: 10.1109/DSD.2019.00029
[72] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “fiction: An Open Source Framework for the Design of Field-coupled Nanocomputing Circuits,” in *International Workshop on Logic Synthesis (IWLS)*, 2019.
[73] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “fiction: Electronic Design Automation for Field-coupled Nanocomputing Circuits,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, **Best Research Demo Award**, 2019.
[74] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “Scalable Design for Field-coupled Nanocomputing Circuits,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2019. DOI: 10.1145/3287624.3287667
[75] R. Wille, **M. Walter**, F. Sill Torres, D. Große, R. Drechsler, “Ignore Clocking Constraints: An Alternative Physical Design Methodology for Field-coupled Nanotechnologies,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2019. DOI: 10.1109/ISVLSI.2019.00121
[76] F. Sill Torres, **M. Walter**, R. Wille, D. Große, R. Drechsler, “Synchronization of Clocked Field-Coupled Circuits,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2018. DOI: 10.1109/NANO.2018.8626294
[77] F. Sill Torres, R. Wille, **M. Walter**, P. Niemann, D. Große, R. Drechsler, “Evaluating the Impact of Interconnections in Quantum-Dot Cellular Automata,” in *Euromicro Conference on Digital System Design (DSD)*, 2018. DOI: 10.1109/DSD.2018.00110
[78] **M. Walter**, R. Wille, D. Große, F. Sill Torres, R. Drechsler, “An Exact Method for Design Exploration of Quantum-dot Cellular Automata,” in *Design, Automation and Test in Europe (DATE)*, 2018.
[79] R. Wille, O. Keszocze, S. Hillmich, **M. Walter**, A. Garcia-Ortiz, “Synthesis of Approximate Coders for On-chip Interconnects Using Reversible Logic,” in *Design, Automation and Test in Europe (DATE)*, 2016. DOI: 10.1109/DATE.2016.7459481
[80] R. Wille, O. Keszocze, **M. Walter**, P. Rohrs, A. Chattopadhyay, R. Drechsler, “Look-ahead Schemes for Nearest Neighbor Optimization of 1D and 2D Quantum Circuits,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2016. DOI: 10.1109/ASPDAC.2016.7428026

Preprints

[81] L. Burgholzer, **M. Walter**, P. Hopf, A. Caride-Tabarés Sánchez, T. Mattsson, B. Hoffmann, N. Färkkilä, D. Bulmash, R. Wille, E. Mansfield, "Practical HPCQC Integration with QDMI: A Real-Hardware Case Study with IQM Systems," 2026.

[82] **M. Walter**, J. Croshaw, S. S. H. Ng, K. Walus, R. Wolkow, R. Wille, "Atomic Defect-Aware Physical Design of Silicon Dangling Bond Logic on the H-Si(100)2x1 Surface," 2023. eprint: 2311.12042.

[83] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, "fiction: An Open Source Framework for the Design of Field-coupled Nanocomputing Circuits," 2019. eprint: 1905.02477.